Penlight

STUDY GUIDE

Pharmacology Basics & Medication Safety

Penlight sample lecture · 11 min lecture · October 17, 2026

Lite mode (offline) Sample content

5	32	47	16
SECTIONS	KEY TERMS	FLASHCARDS	PRACTICE QS

Overview

This lecture (about 11 min, 1,675 words) covers Pharmacokinetics, Vocabulary You Must Know, Heart of Safe Practice and High-alert Medications. The antidote for digoxin toxicity is digoxin immune fab. Naloxone is the antidote for opioid overdose, and the nurse should monitor respiratory rate closely, because the effect of naloxone may wear off before the effect of the opioid.

What you should be able to do

- Explain the main ideas in “Pharmacokinetics”.
- Explain the main ideas in “Vocabulary You Must Know”.
- Explain the main ideas in “Heart of Safe Practice”.
- Explain the main ideas in “High-alert Medications”.
- Explain the main ideas in “Safe Administration Steps for an Intramuscular Injection”.
- Define absorption in your own words.
- Define adverse effect in your own words.
- Define agonist in your own words.
- Recall the priority and safety points the instructor emphasized.

Must-know points

- The therapeutic range is the blood level of a drug that provides the desired effect without causing toxicity. 2:52
- The right documentation means that you document the medication immediately after giving it, never before. 6:11
- Low potassium increases the risk of digoxin toxicity. 9:03
- The antidote for digoxin toxicity is digoxin immune fab. 9:09
- Naloxone is the antidote for opioid overdose, and the nurse should monitor respiratory rate closely, because the effect of naloxone may wear off before the effect of the opioid. 9:20

Lecture notes

1 Pharmacokinetics 0:00

- We will cover pharmacokinetics, the key vocabulary you will see on every exam, the rights of medication administration, and the high-alert medications that need extra attention. 0:10

In both situations the nurse should expect that the provider may lower the dose and should watch closely for signs of toxicity. 1:15

- The first-pass effect refers to the metabolism of an oral drug by the liver before it reaches the general circulation. 1:31
- An antagonist is a drug that binds to a receptor and blocks the effect. 2:09
- Naloxone is a good example of an antagonist because it blocks opioid receptors and reverses an opioid overdose. 2:14

first-pass effect | high-alert medication | liver | rights of medication administration | pharmacokinetics | naloxone

2 Vocabulary You Must Know 2:20

Let's talk about the vocabulary you must know. 2:20

The therapeutic range is the blood level of a drug that provides the desired effect without causing toxicity. 2:52

Toxicity occurs when the drug level rises above the therapeutic range. 3:19

- A side effect is a predictable, often mild, secondary effect of a drug. 3:19
- An idiosyncratic reaction is an unexpected and unusual response to a drug that is different from its known effects. 3:51

therapeutic range | side effect | onset of action | black box warning | trough level | adverse effect

3 Heart of Safe Practice 4:13

- The right patient means using at least two patient identifiers, such as the full name and the date of birth, before giving any medication. 4:34
- The right drug means that you compare the label of the medication with the order on the medication administration record three times: when you take the drug from the storage area, before you prepare it, and at the bedside before you give it. 5:06
- The right dose means that you check the calculation and the dose against the order, and you ask a second nurse to independently verify the dose for high-alert drugs. 5:28
- The right time means giving the drug at the scheduled time. 5:55

The right documentation means that you document the medication immediately after giving it, never before. 6:11

rights of medication administration | right patient | right route | right drug | right dose | right time

4 High-alert Medications 7:37

- Before giving digoxin, check the apical pulse for a full minute. 8:42

> Early signs of digoxin toxicity include nausea, vomiting, anorexia, and visual changes such as yellow-green halos. 8:58

> Low potassium increases the risk of digoxin toxicity. 9:03

> The antidote for digoxin toxicity is digoxin immune fab. 9:09

> Naloxone is the antidote for opioid overdose, and the nurse should monitor respiratory rate closely, because the effect of naloxone may wear off before the effect of the opioid. 9:20

therapeutic range | warfarin | high-alert medication | liver | antidote | naloxone

5 Safe Administration Steps for an Intramuscular Injection 9:30

- Let's talk about safe administration steps for an intramuscular injection. 9:30

> Always check for allergies before giving a new drug. 10:13

> Teach the patient the name of the drug, the purpose, and the side effects to report. 10:19

side effect

Glossary

Absorption	The movement of a drug from the site of administration into the bloodstream.
Adverse effect	An undesired and potentially harmful effect.
Agonist	A drug that binds to a receptor and produces an effect.
Allergic reaction	An immune response to a drug.
Antagonist	A drug that binds to a receptor and blocks the effect.
Antidote	The antidote for digoxin toxicity is digoxin immune fab.
Black box warning	The strongest warning that the Food and Drug Administration requires on a drug label, and it alerts prescribers and nurses about serious risks.
Digoxin	A cardiac glycoside that slows and strengthens the heartbeat.

First-pass effect	The metabolism of an oral drug by the liver before it reaches the general circulation.
Half-life of a drug	The time it takes for the amount of drug in the body to decrease by half.
Heparin	An anticoagulant that is monitored with the activated partial thromboplastin time, or aPTT.
High-alert medication	High-alert medications are drugs that carry a heightened risk of causing significant patient harm when used in error.
Idiosyncratic reaction	An unexpected and unusual response to a drug that is different from its known effects.
Liver	Remember that liver and kidney function matter a great deal.
Medication reconciliation	The process of comparing the medications a patient is taking at home with the medications ordered in the hospital, in order to prevent duplications, omissions, and interactions.
Naloxone	A good example of an antagonist because it blocks opioid receptors and reverses an opioid overdose.
Onset of action	The time it takes for a drug to begin working.
Opioid	Naloxone is the antidote for opioid overdose, and the nurse should monitor respiratory rate closely, because the effect of naloxone may wear off before the effect of the opioid.
Pharmacodynamics	The study of what a drug does to the body.
Pharmacokinetics	The study of what the body does to a drug.
Right documentation	It means that you document the medication immediately after giving it, never before.
Right dose	It means that you check the calculation and the dose against the order, and you ask a second nurse to independently verify the dose for high-alert drugs.
Right drug	It means that you compare the label of the medication with the order on the medication administration record three times: when you take the drug from the storage area, before you prepare it, and at the bedside before you give it.

Right patient	It means using at least two patient identifiers, such as the full name and the date of birth, before giving any medication.
Right reason	It means that you know why the patient is receiving the drug and that it fits the patient's condition.
Right response	It means that you evaluate whether the drug had the intended effect and whether the patient had any adverse reaction.
Right route	It means that the drug is given by the route that was ordered, and that the route is safe for the patient.
Right time	It means giving the drug at the scheduled time.
Side effect	A predictable, often mild, secondary effect of a drug.
Therapeutic range	The blood level of a drug that provides the desired effect without causing toxicity.
Trough level	The lowest concentration of a drug in the blood, and it is drawn just before the next dose is given.
Warfarin	An oral anticoagulant that is monitored with the international normalized ratio, or INR.

Quick review checklist

- [ ] The therapeutic range is the blood level of a drug that provides the desired effect without causing toxicity.
- [ ] The right documentation means that you document the medication immediately after giving it, never before.
- [ ] Low potassium increases the risk of digoxin toxicity.
- [ ] The antidote for digoxin toxicity is digoxin immune fab.
- [ ] Naloxone is the antidote for opioid overdose, and the nurse should monitor respiratory rate closely…
- [ ] We will cover pharmacokinetics, the key vocabulary you will see on every exam, the rights of medication administration, and the high-alert medications that need extra attention.
- [ ] Let's talk about the vocabulary you must know.
- [ ] The right patient means using at least two patient identifiers, such as the full name and the date of birth, before giving any medication.

Flashcard preview

Absorption
The movement of a drug from the site of administration into the bloodstream.

Adverse effect
An undesired and potentially harmful effect.

Agonist
A drug that binds to a receptor and produces an effect.

Allergic reaction
An immune response to a drug.

Antagonist
A drug that binds to a receptor and blocks the effect.

Black box warning
The strongest warning that the Food and Drug Administration requires on a drug label, and it alerts prescribers and nurses about serious risks.

Digoxin
A cardiac glycoside that slows and strengthens the heartbeat.

First-pass effect
The metabolism of an oral drug by the liver before it reaches the general circulation.

Half-life of a drug
The time it takes for the amount of drug in the body to decrease by half.

Heparin
An anticoagulant that is monitored with the activated partial thromboplastin time, or aPTT.

Self-test

1. Which term is described here? “The study of what a drug does to the body.”

A. Pharmacodynamics
B. Right response
C. Antidote
D. Digoxin toxicity

2. Which term is described here? “A drug that binds to a receptor and produces an effect.”

A. Agonist
B. Right drug
C. Allergic reaction
D. Pharmacokinetics

3. Which term is described here? “The metabolism of an oral drug by the liver before it reaches the general circulation.”

A. Antagonist
B. Right dose
C. Heparin
D. First-pass effect

4. Which term is described here? “The lowest concentration of a drug in the blood, and it is drawn just before the next dose is given.”

A. First-pass effect
B. Antidote
C. Right response
D. Trough level

5. Which term is described here? “A drug that binds to a receptor and blocks the effect.”

A. Right time
B. Antagonist
C. High-alert medication
D. Vitamin K

6. Which term is described here? “An anticoagulant that is monitored with the activated partial thromboplastin time, or aPTT.”

A. Agonist
B. Onset of action
C. Heparin
D. Right response

7. Which term is described here? “The process of comparing the medications a patient is taking at home with the medications ordered in the hospital, in order to prevent duplications, omissions, and interactions.”

A. Opioid
B. Warfarin
C. Medication reconciliation
D. Adverse effect

8. According to the lecture, what is Digoxin?

A. A predictable, often mild, secondary effect of a drug.
B. A good example of an antagonist because it blocks opioid receptors and reverses an opioid overdose.
C. A cardiac glycoside that slows and strengthens the heartbeat.
D. It means using at least two patient identifiers, such as the full name and the date of birth, before giving any medication.

Answer key

1. A — The lecture defines Pharmacodynamics: “The study of what a drug does to the body.” (≈1:58)

2. A — The lecture defines Agonist: “A drug that binds to a receptor and produces an effect.” (≈2:03)

3. D — The lecture defines First-pass effect: “The metabolism of an oral drug by the liver before it reaches the general circulation.” (≈1:31)

4. D — The lecture defines Trough level: “The lowest concentration of a drug in the blood, and it is drawn just before the next dose is given.” (≈3:03)

5. B — The lecture defines Antagonist: “A drug that binds to a receptor and blocks the effect.” (≈2:09)

6. C — The lecture defines Heparin: “An anticoagulant that is monitored with the activated partial thromboplastin time, or aPTT.” (≈8:04)

7. C — The lecture defines Medication reconciliation: “The process of comparing the medications a patient is taking at home with the medications ordered in the hospital, in order to prevent duplications, omissions, and interactions.” (≈7:21)

8. C — “Digoxin” — A cardiac glycoside that slows and strengthens the heartbeat. (≈8:36)

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